

UNIT 2

1 Mark Questions

1. What does IPR stand for in Cyber Laws?

→ *Intellectual Property Rights.*

2. Expand COBIT in the context of security governance.

→ *Control Objectives for Information and Related Technology.*

3. Define Security Assurance in one sentence.

→ *It is the confidence that security measures protect systems and data as intended.*

4. Mention any one international standard for information security.

→ *ISO/IEC 27001.*

5. What is the primary objective of a Security Audit?

→ *To evaluate and verify the effectiveness of security controls.*

2 Mark Questions

1. Define Security Assurance in the context of cyber laws.

- *Security Assurance* is the process of producing evidence (through testing, evaluation and review) that security controls are correctly implemented and effective in protecting systems and data to meet legal, regulatory and organizational requirements.
 - *Example:* Pen-tests and compliance checks that demonstrate adherence to a data-protection law.
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2. What is Intellectual Property Rights (IPR)? Give one example.

- *Intellectual Property Rights (IPR)* are legal rights that protect creations of the mind (e.g., inventions, literary and artistic works, designs, trademarks), giving creators exclusive control over use and distribution for a limited time.
 - *Example:* Copyright on software source code.
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3. Name any two international standards used in cybersecurity.

- *ISO/IEC 27001* — standard for Information Security Management Systems (ISMS) requirements.
 - *ISO/IEC 27002* — code of practice that provides guidelines for implementing information security controls.
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4. What is the role of a security audit?

- The role of a *security audit* is to systematically assess security policies, controls and procedures to verify compliance, identify weaknesses or gaps, and provide recommendations to reduce risk and improve security posture.
 - *Example:* An internal audit that verifies controls required for ISO/IEC 27001 certification.
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5. Expand COBIT and briefly state its purpose.

- **COBIT** — Control Objectives for Information and Related Technology.
 - *Purpose:* A governance and management framework that helps organizations align IT processes with business goals, define control objectives, assign responsibilities, and measure IT performance and compliance.
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4 Mark Questions

1. Explain the difference between Security Assurance and Security Compliance with suitable examples.

- **Security Assurance** is the process of building and providing objective evidence that security controls are correctly designed, implemented, and effective in reducing risks — it's about *confidence* in security. Methods include threat modelling, penetration testing, formal verification, and continuous monitoring.
- **Security Compliance** means meeting specific legal, regulatory or standards-based requirements (e.g., GDPR, ISO/IEC 27001, PCI-DSS); it is *conformance* to externally defined rules or checklists. Compliance shows that minimum controls are in place, but passing compliance checks does not guarantee real security.

- *Example:* A company may be **compliant** with ISO/IEC 27001 (has documented controls and audit reports) but still improve **assurance** by running red-team exercises and continuous monitoring to demonstrate real-world effectiveness.
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2. Describe the importance of cyber laws in preventing digital crimes.

- Cyber laws create the legal definitions, penalties, and procedures needed to deter, investigate, and punish digital offences (e.g., hacking, identity theft, data breaches), thereby raising the cost of cybercrime.
 - They enable lawful evidence collection, cross-border cooperation (MLATs and mutual legal assistance), takedown of illicit content, and remedies for victims — all of which make prosecution and remediation feasible.
 - By setting obligations for organizations (data protection, breach notification, liability), cyber laws also push businesses to adopt better security practices and accountability, reducing the overall risk of digital crimes.
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3. What are the major components of an effective Security Audit process?

- **Scope s Objectives:** Define assets, systems, standards, and audit goals (what's covered and why).
 - **Planning s Risk Assessment:** Identify critical systems, threats, and controls to prioritize audit activities.
 - **Evidence Collection s Testing:** Inspect configurations, logs, policies, and run tests (vulnerability scans, control checks, interviews).
 - **Evaluation s Findings:** Compare evidence against standards/policies, identify gaps and risk severity.
 - **Reporting s Recommendations:** Produce a clear report with prioritized remediation actions, timelines, and owners.
 - **Remediation Follow-up s Closure:** Verify fixes, re-test, and document closure; maintain audit trail for compliance and continuous improvement.
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4. Write a short note on SSE-CMM and its application in cybersecurity.

- **SSE-CMM (Systems Security Engineering Capability Maturity Model)** is a process reference/maturity model for security engineering that defines best practices and capability levels for organizations performing security engineering. Its goal is to make security engineering a measurable, repeatable discipline by describing process areas and maturity levels.
 - The model helps organizations assess and improve how they plan, design, implement, verify, and manage security across the system lifecycle; it is used to choose qualified providers, guide process improvement, and augment assurance activities.
 - **Application:** SSE-CMM is applied to evaluate security engineering capability (e.g., secure requirements, risk analysis, verification), guide investments in training/tools, and raise process maturity so that security becomes systematic rather than ad-hoc
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5. Discuss IPR violations in cyberspace with two relevant examples.

- **Copyright infringement / Piracy:** Unauthorized copying, distribution or streaming of copyrighted works (software, movies, music, e-books) via file-sharing sites or torrent networks. This harms creators and businesses; remedies include DMCA takedowns, civil suits, and criminal penalties in some jurisdictions. *Example:* Uploading and distributing pirated software or movies on a torrent site; the site operator and uploaders can be subject to takedown requests and legal action.
 - **Trademark abuse and counterfeiting / Domain squatting:** Using identical or confusingly similar brand names, images, or domain names to sell counterfeit goods or to mislead customers (phishing/typosquatting). This damages brand reputation and misleads consumers. *Example:* A fake e-commerce listing that uses a well-known brand's trademark to sell counterfeits, or registering a domain like "amaz0n-payments.com" to phish user credentials — both actionable under IPR and consumer-protection laws.
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6 Mark Questions

1. Explain the objectives and features of the COBIT framework. How does it help in IT governance and cybersecurity?

COBIT (Control Objectives for Information and Related Technology) is an internationally recognized IT governance and management framework developed by **ISACA**. It provides organizations with a structured approach to align IT operations with business goals, manage risks, and ensure compliance with regulatory standards.

The **main objectives** of COBIT are to:

- Establish a common language between business and IT management.
- Ensure that IT processes are efficient, reliable, and aligned with organizational strategy.
- Facilitate compliance with laws, regulations, and industry standards.
- Measure and improve the performance and maturity of IT controls.

Key features include:

- **Comprehensive Process Framework:** Divides governance and management into domains such as *Plan and Organize*, *Acquire and Implement*, *Deliver and Support*, and *Monitor and Evaluate*.
- **Performance Measurement Tools:** Uses maturity models and KPIs to assess control effectiveness.
- **Risk and Compliance Integration:** Helps identify, assess, and mitigate IT-related risks.
- **Alignment with Standards:** Integrates with frameworks like ISO/IEC 27001 and ITIL for cohesive governance.

In **cybersecurity**, COBIT enhances accountability, ensures data integrity, and establishes consistent control mechanisms. It provides a measurable way to evaluate how well security objectives are achieved.

Example: A financial institution applying COBIT can ensure secure transaction processing, regulatory compliance, and reduced data breach risks through continuous governance monitoring.

2. Describe the different types of Intellectual Property Rights (IPR) and their importance in the digital world.

Intellectual Property Rights (IPR) are the legal rights that protect creations of the human mind — such as inventions, literary works, designs, and symbols — ensuring that creators receive recognition and economic benefits from their innovations. These rights are vital in fostering innovation and fair competition, especially in the **digital age**, where duplication and misuse are common.

The **major types of IPR** include:

- **Copyright:** Protects original creative works such as software programs, e-books, music, and videos.
- **Patents:** Granted for new inventions, processes, or technological improvements that provide a unique function or advantage.
- **Trademarks:** Safeguard brand identifiers like logos, names, and symbols used in commerce to distinguish products or services.
- **Design Rights:** Protect the visual or aesthetic aspects of products (e.g., icons, UI designs).
- **Trade Secrets:** Cover confidential business information, formulas, or algorithms not disclosed publicly.

In today's **digital ecosystem**, IPR ensures that content creators, developers, and companies are protected from theft, piracy, and counterfeiting. It encourages research, innovation, and fair competition in global markets.

Example: Apple's patented design of the iPhone and its trademarked logo protect it from unauthorized copies, while copyright laws safeguard its iOS software code.

3. Discuss the process of conducting a security audit in an organization. What tools and standards are generally followed?

A **security audit** is a systematic evaluation of an organization's information systems to determine how effectively security policies, procedures, and controls are implemented. It identifies vulnerabilities, ensures compliance with standards, and recommends corrective measures to strengthen the security posture.

Process of Security Audit:

1. **Planning and Scope Definition:** Identify audit objectives, assets to be tested, and applicable regulations.
2. **Information Gathering:** Collect data about systems, networks, and existing controls.
3. **Risk Assessment:** Evaluate threats and potential impacts on assets.
4. **Testing and Evaluation:** Perform vulnerability scanning, penetration testing, and configuration reviews.
5. **Reporting:** Document findings, classify risks, and suggest mitigation steps.
6. **Follow-up:** Verify corrective actions and ensure continuous improvement.

Common Tools Used:

- *Nmap* (network scanning), *Nessus* (vulnerability assessment), *Wireshark* (traffic analysis), *Metasploit* (penetration testing).

Standards and Frameworks Followed:

- **ISO/IEC 27001, COBIT, NIST SP 800-53, and PCI-DSS** for financial systems.

Security audits ensure that IT infrastructure remains compliant and resilient against evolving threats.

Example: Banks conduct regular ISO 27001-based audits to detect configuration weaknesses and prevent unauthorized data access.

4. What is SSE-CMM (Systems Security Engineering Capability Maturity Model)? Explain its structure and relevance in digital forensics.

The **Systems Security Engineering Capability Maturity Model (SSE-CMM)** is a framework developed to evaluate and improve an organization's ability to engineer and manage secure systems throughout their lifecycle. It helps organizations assess the maturity and consistency of their security engineering processes.

Structure of SSE-CMM:

- **Domains:** Divided into 22 process areas such as risk management, security testing, configuration management, and incident response.
- **Maturity Levels:** Ranges from *Level 1 (Performed Informally)* to *Level 5 (Continuously Improving)*, indicating process sophistication.
- **Process Improvement:** Each level defines measurable goals, practices, and performance indicators for continual enhancement.

Relevance in Digital Forensics:

- Ensures that forensic tools and procedures are developed and maintained systematically.
- Promotes reliable, repeatable, and legally defensible investigations.
- Encourages continuous improvement in evidence handling and analysis methods.

SSE-CMM thus bridges the gap between **engineering practices** and **security assurance** in forensic environments.

Example: Government defense agencies use SSE-CMM to validate that their digital forensic labs operate under consistent, auditable processes that maintain data integrity and confidentiality.

5. Evaluate the role of international security standards like ISO/IEC 27001 in ensuring data protection and global compliance.

ISO/IEC 27001 is an internationally recognized standard for establishing, implementing, maintaining, and continually improving an **Information Security Management System (ISMS)**. It helps organizations safeguard sensitive data and comply with global regulatory requirements.

Key Roles in Data Protection and Compliance:

1. **Systematic Risk Management:** Identifies, analyzes, and mitigates data security risks across people, processes, and technology.
2. **Access Control and Data Integrity:** Ensures that only authorized users access information and that data remains accurate and consistent.
3. **Policy Standardization:** Provides a globally accepted framework for developing and enforcing security policies.
4. **Legal and Regulatory Alignment:** Assists in compliance with GDPR, HIPAA, and other data protection laws.
5. **Continuous Improvement:** Mandates regular audits and reviews to adapt to emerging cyber threats.

By adopting ISO/IEC 27001, organizations demonstrate accountability and trustworthiness, which enhances customer confidence and global interoperability.

Example: Tech giants like Microsoft and Google maintain ISO 27001 certification to guarantee secure data handling for their cloud services, ensuring compliance with international privacy standards.

UNIT 3

1 Mark Questions

1. Define Computer Forensics.

→ *Computer Forensics is the process of collecting, preserving, analyzing, and presenting digital evidence in a legally acceptable manner.*

2. Expand FAT and NTFS.

→ *FAT - File Allocation Table; NTFS - New Technology File System.*

3. Name any two UNIX file systems.

→ *UFS (Unix File System), ext (Extended File System).*

4. What is meant by “Data Acquisition” in forensics?

→ *It is the process of collecting and imaging digital evidence from a device while ensuring its integrity.*

5. Give one example of an Internet Artifact.

→ *Browser history.*

2 Mark Questions

1. State any two cardinal rules of computer forensics.

- **Preserve the original evidence:** Never alter the original media; always create a verified forensic image/copy for analysis.
- **Maintain chain of custody s document everything:** Record who handled the evidence, when, where and what was done to it to ensure admissibility in court.

2. Differentiate between FAT16 and FAT32.

Feature	FAT16	FAT32
Designed for / Introduced	Older FAT variant used on small volumes and early PCs.	Introduced later to support much larger disks and improve space efficiency.
Volume s file size (practical)	Suitable for smaller partitions; limited maximum volume/file sizes.	Supports much larger partitions and files (designed for modern larger disks).
Cluster size s space efficiency	Uses larger clusters on bigger volumes, causing more internal slack (wasted space).	Uses smaller clusters for large volumes, improving storage efficiency.
Root directory s structure	Fixed-size root directory (limited number of entries).	Root directory is a normal chainable cluster—no fixed-entry limit.

3. What is the purpose of authentication in the forensic process? Authentication ensures that digital evidence is genuine, untampered, and legally admissible. It provides confidence that the data collected is an exact copy of the original and has not been altered during acquisition, transfer, or analysis. This is usually achieved using cryptographic hash values (like MD5 or SHA) which remain constant if the evidence is unchanged.

4. List two examples of computer artifacts.

- **Windows Registry keys** (e.g., MRU lists, user profile info).
 - **System/event logs** (e.g., Windows Event Log, syslog entries).
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5. What is the significance of OS artifacts in forensic investigation? OS artifacts are traces left behind by the operating system, such as system logs, registry keys, temp files, prefetch data, or recent file lists. These artifacts are significant because they help investigators reconstruct user actions, verify timelines, detect unauthorized access, and identify which programs or files were recently used. They provide context that supports other evidence and are often crucial in linking suspects to activities on a device.

4 Mark Questions

1. Explain the process of data acquisition in computer forensics.

- **Authorization & preparation:** Obtain legal authority (warrant/consent), define scope and target devices, and prepare tools (write-blockers, imaging software).
- **Secure & document scene:** Photograph devices, label and log their state (powered on/off, network connections) and maintain chain of custody.
- **Collect volatile data (if live):** If the system is running, capture RAM, network connections, and running processes before shutdown.
- **Create forensic images:** Use a write-blocker to create bit-for-bit images (dd/FTK Imager/Guymager) of drives or partitions and capture copies of relevant removable media.
- **Authenticate & store:** Compute and record cryptographic hashes (MD5/SHA-1/SHA-256) of originals and images, verify hashes post-copy, and store evidence in secure, access-controlled storage with full documentation. *Example:* Capture RAM with a live tool, then image the disk through a hardware write-blocker and record SHA-256 hashes for both original and image.

2. Compare FAT file system with NTFS with respect to security and efficiency.

- **Security (access control s integrity):** FAT has no built-in ACLs, encryption or journaling — minimal security; NTFS supports ACLs, file encryption (EFS), auditing and journaling (transactional metadata), so it is far more secure and forensically informative.
- **Metadata s recoverability:** FAT stores simple directory entries and a File Allocation Table (less metadata), making some carving easier but less contextual info; NTFS stores rich metadata in the MFT (timestamps, ACLs, alternate data streams), aiding attribution and timeline reconstruction.
- **Fault tolerance s recovery:** FAT (no journal) is more prone to corruption on crashes; NTFS journaling improves consistency and speeds safe recovery, though the journal can overwrite quick recovery traces.
- **Efficiency s scalability:** FAT is simple and efficient on small volumes/removable media but wastes space with large clusters on big volumes; NTFS scales to larger disks and files with better space management and performance for many small files.

Example: USB sticks often use FAT32 for cross-platform compatibility, while NTFS is preferred for Windows system disks because of security and metadata.

3. Describe the importance of Internet artifacts in forensic investigations.

- **Definition s sources:** Internet artifacts are browser and network traces (history, cache, cookies, bookmarks, download records, DNS and proxy logs, server logs, email headers) left by web activity.
- **Reconstructing activity s intent:** They provide direct evidence of visited sites, searches, downloads, and timestamps — useful to reconstruct what a user did online and whether actions were deliberate (intent).
- **Linking local to remote:** Cookies, session tokens and cached credentials can link a local machine to specific online accounts or server sessions; server logs and ISP records can tie activity to IP addresses and timeframes.
- **Evidence of exfiltration s communications:** Cache and download entries show transferred files, while network logs can reveal data exfiltration channels or communications with command-and-control servers.
Example: Browser history + server logs can show a user accessed and downloaded illicit material at a given time; cookie/session data can corroborate an authenticated session.

4. Write short notes on UNIX file systems used in forensics.

- **ext2 / ext3 / ext4 (Linux):** ext2 (no journal) allows straightforward carving; ext3 adds journaling (safer on crash but journal affects recoverability); ext4 adds extents and larger volume/file support — forensic focus is on inodes (metadata), timestamps (atime/mtime/ctime), and journal handling which affects deleted-file recovery.
 - **UFS / FFS (BSD, older UNIX):** Uses inodes and cylinder groups; file metadata lives in inodes and deletion often clears directory entries but leaves blocks until overwritten — forensic tools inspect inode tables and indirect blocks.
 - **XFS / ReiserFS:** High-performance, journaled systems used on servers (XFS), with allocation groups and advanced allocation strategies; journaling improves integrity but complicates recovery of recent deletes.
 - **Forensic implications (common):** Investigators analyze inodes, indirect blocks, and journals; they use specialized tools (e.g., The Sleuth Kit, debugfs) to extract metadata, recover deleted files, and build timelines.
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5. Explain any two forensic applications of OS artifacts with examples.

- Timeline reconstruction:** OS artifacts like file system timestamps (MFT entries on NTFS / inodes on UNIX), Windows Event Logs, macOS Unified Logs and prefetch/journal entries allow investigators to build a precise timeline of file creation, modification, execution and system events.
Example: MFT timestamps + Windows Event Log entries can show when a malicious executable was copied and later executed, establishing sequence and timeframe.
- User activity s device attribution:** Artifacts such as Windows Registry MRU lists, USBSTOR registry keys, shellbags, and mount/last-seen entries record recently used files, attached devices and folders accessed, helping to attribute actions to a user or link a removable device by serial number.
Example: USBSTOR registry entries revealing a USB drive's serial number matched to a physical drive seized from a suspect links that device to files found on the system.

6 Mark Questions

1. Discuss in detail the cardinal rules of computer forensics (with examples)

Definition / Purpose: Cardinal rules are core principles that ensure digital evidence is collected, handled, analyzed, and presented so it remains reliable, admissible, and defensible in court.

Key rules (with brief explanation):

- **Preserve the original evidence:** Never work on the original media; create a bit-for-bit forensic image using a hardware or software write-blocker to avoid modification. *Example:* Forensically image a suspect HDD to an evidence drive and compute hashes of both.
- **Maintain chain of custody:** Log every transfer, handler, date/time, and action so the evidence trail is auditable. *Example:* A signed log showing who transported the drive from scene to lab.
- **Authenticate and verify:** Use cryptographic hashes (MD5/SHA-1/ SHA-256) before and after copying to prove integrity.
- **Document everything:** Record tools, versions, commands, settings, and analysis steps to ensure reproducibility.
- **Collect volatile data first:** Capture RAM, running network connections, and live processes before powering down systems, because these are ephemeral.
- **Use validated tools & repeatable methods:** Rely on accepted forensic tools and standard procedures to reduce challenges in court.
- **Avoid scope creep & respect legal limits:** Follow warrants and legal authority; do not access unrelated data.

Conclusion: Following these rules preserves probative value, supports expert testimony, and reduces the risk of evidence exclusion.

2. Explain Windows file systems (FAT12, FAT16, FAT32, NTFS) and their forensic importance

Brief descriptions:

- **FAT12:** Early FAT used on floppy disks; very small cluster and table sizes.
- **FAT16:** Older DOS/early Windows file system supporting modest partition sizes; root directory fixed in location and size.

- **FAT32:** Improved FAT allowing much larger partitions and more clusters; root directory is cluster-based.
- **NTFS:** Modern Windows file system with Master File Table (MFT), journaling (\$LogFile), metadata attributes, ACLs, alternate data streams (ADS), compression, and encryption support.

Forensic importance & artifacts:

- **File allocation & deleted files:** In FAT systems, deleted entries often mark first char with 0xE5 and cluster chains remain until overwritten—enabling file carving but with limited metadata. In NTFS, the MFT retains rich metadata (timestamps, file size, attributes) and often holds records after deletion, improving recovery chances.
- **Timestamps & timelines:** NTFS stores multiple timestamps (created, modified, MFT changed, accessed) useful for timeline reconstruction. FAT has fewer timestamp details.
- **Slack space & unallocated clusters:** Both FSs can contain remnants of prior files; slack analysis can reveal fragments.
- **Journaling & logs:** NTFS \$LogFile and USN Journal record filesystem changes useful to prove actions and sequence.
- **ADS & EFS:** NTFS features (ADS, Encrypting File System) can hide data or affect access — important to check in investigations.

Example: Recovering a deleted document: on FAT32 you may recover raw content if clusters persist; on NTFS you can often recover file metadata from the MFT even if content blocks are partially overwritten, aiding attribution and timeline.

3. Describe the authentication process of electronic evidence in forensic science.

Authentication of electronic evidence is the process of verifying that the collected digital data is genuine, untampered, and identical to its original source. It ensures the integrity and admissibility of evidence in legal proceedings.

Steps in the Authentication Process:

- **Identification:** Recognize and label potential sources of digital evidence such as hard drives, USBs, or network logs.
- **Collection:** Acquire data using forensically sound methods like bit-stream imaging, ensuring the original media is not altered.

- **Hashing and Integrity Verification:** Generate cryptographic hash values (MD5, SHA-256) for both the original and copied data. Matching hashes confirm authenticity.
- **Chain of Custody:** Maintain a complete record of who handled the evidence, when, and for what purpose to establish accountability.
- **Documentation:** Record tools, versions, time stamps, and system settings for reproducibility.
- **Validation:** Use recognized forensic tools (e.g., EnCase, FTK) and repeat tests to confirm results.

Importance: Authentication builds legal credibility and prevents challenges of evidence manipulation.

Example: When imaging a suspect's laptop, investigators compute a SHA-256 hash before and after imaging; identical hashes verify that no data was modified during acquisition.

4. Explain different types of artifacts (Computer, Internet, OS) and their forensic applications.

Artifacts are residual traces or data fragments left behind by user actions, applications, or the operating system. They are critical in reconstructing user activity and system behavior during an investigation.

Types of Artifacts:

- **Computer Artifacts:** Include files, logs, registry entries, and system configuration data that show system usage.
Example: Prefetch files in Windows reveal which programs were recently executed.
- **Internet Artifacts:** Data generated by browsers such as history, cookies, cache, downloads, and bookmarks.
Example: Browser cache files can confirm that a suspect visited specific websites or downloaded certain files.
- **OS Artifacts:** System-level data like event logs, pagefiles, hibernation files, and registry hives.
Example: Windows event logs record logon attempts, shutdowns, and USB connections.

Forensic Applications:

Artifacts help investigators determine:

- User identities and login timelines
- File access and deletion patterns
- Communication or browsing history
- Presence of malicious or hidden activity

Example: In a cyber-fraud case, investigators used browser history and registry entries (artifacts) to confirm that fraudulent transactions originated from the suspect's PC.

5. Compare and contrast Windows, UNIX, and Mac file systems from a forensic perspective.

Aspect	Windows (NTFS/FAT)	UNIX (ext3/ext4, UFS)	Mac (HFS+, APFS)
Structure	Uses Master File Table (MFT) with metadata for each file; supports journaling, permissions, and Alternate Data Streams.	Uses inodes to store file metadata; supports journaling and permission bits.	Uses Catalog File and Allocation Blocks (HFS+) or Containers and Volumes (APFS) with snapshots.
Metadata & Timestamps	Multiple timestamps (created, modified, accessed, MFT changed).	Maintains access, modification, and change timestamps.	Stores extended timestamps and supports cloning for version tracking.
Deleted File Recovery	MFT retains deleted entries until overwritten; easier recovery.	Data blocks often reused quickly, making recovery difficult.	APFS snapshots can assist recovery; otherwise metadata-intensive.
Security Features	ACLs, EFS encryption, ADS for hidden data.	File permissions and ownership model.	FileVault encryption, sandboxing.
Forensic Tools	EnCase, FTK, Autopsy, X-Ways.	Sleuth Kit, Autopsy, dd.	MacQuisition, BlackLight, Cellebrite.

Summary:

Windows offers rich metadata and easier recovery due to NTFS structure. UNIX provides robust access control but limited recovery once overwritten. Mac systems, with APFS snapshots and encryption, require specialized tools.

Example: In a corporate leak case, analysts used NTFS MFT records to confirm when a confidential document was accessed and copied to an external USB drive.